

BIOLOGY — SEMESTER - I

3 lesson(s) included, 1 missing

Included:

- Chapter 1 Plant life: The Leaf.
- Chapter 2 Plant life: The Flower
- Chapter 3 Cell : The basic unit of life

MISSING (no downloaded Question Bank found):

- Chapter 4 Human Body : Digestive System

Question 1.

I. Choose the correct option for the following assertion and reasoning questions.

(a) Both A and R are true and R explains A

(b) Both true but R not explanation

(c) A true, R false

(d) A false, R true

1. Assertion (A): Leaves are the food factories of plants.

Reason (R): Leaves prepare food by photosynthesis.

2. Assertion (A): Transpiration takes place in leaves.

Reason (R): Leaves have tiny pores called stomata.

3. Assertion (A): All leaves are similar in shape and size.

Reason (R): Leaves perform the same function in all plants.

4. Assertion (A): Veins are present in the lamina of the leaf.

Reason (R): Veins help in transport of water and food.

5. Assertion (A): Reticulate venation is found in dicot plants.

Reason (R): Parallel venation is found in monocot plants.

6. Assertion (A): Leaves help in gaseous exchange.

Reason (R): Carbon dioxide enters through stomata.

7. Assertion (A): Petiole connects the leaf to the stem.

Reason (R): Petiole supports the lamina.

8. Assertion (A): Leaves lose water during transpiration.

Reason (R): Stomata remain completely closed at all times.

9. Assertion (A): Some leaves are modified into spines.

Reason (R): This helps in reducing water loss.

10. Assertion (A): All leaves are green in colour.

Reason (R): Leaves contain chlorophyll.

II. Think and answer.

1. A plant is kept inside a closed room for many days without sunlight. Its leaves turn pale. Explain the reason.
2. A student applies oil on both sides of a leaf. What changes will occur in transpiration and gas exchange? Explain.
3. Why do desert plants like cactus survive even with very little water?
4. During heavy rainfall, some plants show drooping leaves. Explain how transpiration and water balance are involved.
5. Why do aquatic plants have large and broad leaves compared to desert plants?
6. If a plant has damaged veins in its leaves, how will it affect the plant's survival? Explain.
7. Two plants are kept under the same conditions. One has broad leaves and the other has needle-like leaves. Which one will lose more water and why?
8. A farmer notices that crops with damaged leaves grow poorly. Explain the reason based on leaf functions.
9. Why are stomata more important during daytime than at night?
10. A plant shows very slow growth even though it is watered regularly. What could be the role of leaves in this condition?

III. Answer the following questions.

1. Lamina (Shape and function)
2. Petiole (Function)
3. Stomata (Location and function)
4. Cactus – Modified leaf (Function)
5. Leaf margin. (Location)

IV. Fill in the blanks.

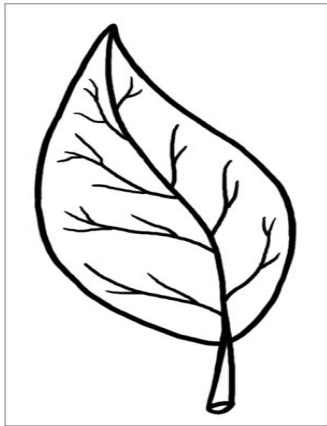
1. The flat, green part of the leaf is called the _____.
2. The stalk that attaches the leaf to the stem is called the _____.
3. The central thick vein of a leaf is called the _____.
4. The network of veins in a leaf is known as _____.
5. The tiny pores present on the leaf surface are called _____.
6. Leaves prepare food by the process of _____.
7. The loss of water from leaves is called _____.
8. In monocot plants, the venation is _____.

9. In dicot plants, the venation is _____.
10. Leaves of cactus are modified into _____.
11. The green colour of leaves is due to the presence of _____.
12. The exchange of gases in leaves occurs through _____.
13. The edge of the leaf is called the leaf _____.
14. The tip of the leaf is called the leaf _____.
15. Veins help in the transport of _____ and _____.

Question 2

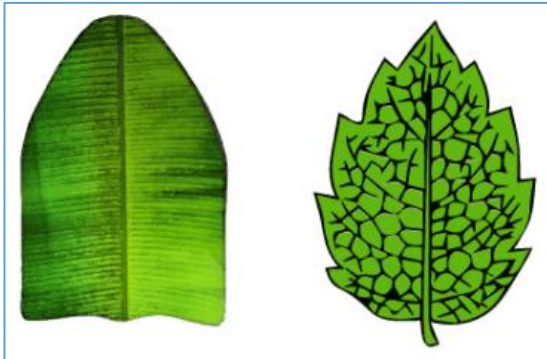
I. Picture based questions

1.



- i) Identify the picture
- ii) Label all the parts in the picture
- iii) Write the function of the given picture.
- iv) Define
 - a) Photosynthesis
 - b) Transpiration

2.

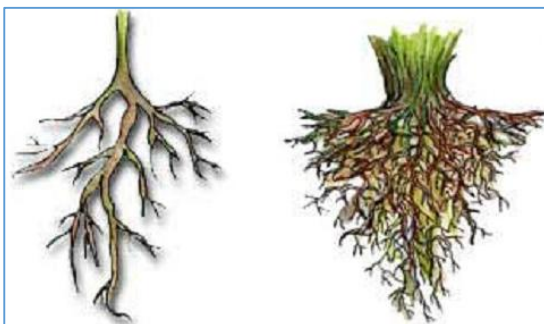


Pic A

Pic B

- i) Identify the types of leaf in picture A and B
- ii) Observe the shape of the leaves and identify which belongs to monocot and dicot plant
- iii) Give an example for picture A and B

3.



Pic A

Pic B

- i) Identify the types of root from the picture A and B
- ii) Differentiate Picture A and B.
- iii) Write any two functions of root.

4.



i) Name the leaf

ii) Define vegetative propagation

iii) In which part of the leaf does the adventitious bud grow?

II. Draw and label the following.

1. Parts of a typical leaf
2. Compound leaf
3. Parts of a shoot system
4. Adventitious buds on a bryophyllum leaf.

Question 1.

I. Choose the correct option for the following assertion and reasoning questions.

(a) Both A and R are true and R explains A

(b) Both true but R not explanation

(c) A true, R false

(d) A false, R true

1. Assertion (A): Sepals are leaf like structures in calyx

Reason (R): They are green and prepare food.

2. Assertion (A): Each stamen has an anther at its tip.

Reason (R): Anthers contain the female gametes.

3. Assertion (A): Wind pollinated flowers are not colourful.

Reason (R): Sunflower plant is pollinated by wind.

4. Assertion (A): Vallisneria is pollinated by wind.

Reason (R): It is a bisexual plant.

5. Assertion (A): Garden Pea is an example of fleshy fruit.

Reason (R): The pericarp develops the succulent tissues.

6. Assertion (A): All fruits contain many seeds.

Reason (R): Fruits develop from flowers.

7. Assertion (A): Water is essential for germination.

Reason (R): It helps the seed to absorb nutrients.

8. Assertion (A): Seeds need air to grow.

Reason (R): The embryo requires oxygen for respiration.

9. Assertion (A): Animals help in seed dispersal.

Reason (R): Seeds stick to the animal fur or eaten.

10. Assertion (A): Radical is the embryonic root.

Reason (R): It gives rise to the root system.

II. Think and answer.

1. Why is pollination considered an important step before fertilization. Give reason.
2. Why are flowers called the reproductive organs of plants?
3. A farmer notices fewer bees in his field. How might this affect fruit production?
4. What would happen if pollen grains could not reach the stigma?
5. Why do gardeners encourage bees and butterflies to visit flowering plants?
6. Why are most flowers found at the tip of branches rather than underground?
7. A flower has petals but no stamens or pistil. Can it produce seeds? Give a reason.
8. How would a plant be affected if it could not produce flowers?
9. Why is it important that the stigma is often sticky?
10. The ovary becomes a fruit after fertilization. How does this benefit the seeds?

III. Answer the following questions.

1. Stigma (Shape and function)
2. Endocarp (Function)
3. Why do farmers usually sow seeds during a particular season rather than throughout the year?
4. Differentiate
a) self pollination and cross pollination b) monocot and dicot seeds
c) complete and incomplete flowers d) dry and fleshy fruits
5. Define fertilization.
6. A seedling fails to develop properly because its cotyledons were damaged before germination.
How might this affect the young plant? What does this tell you about the importance of cotyledons?

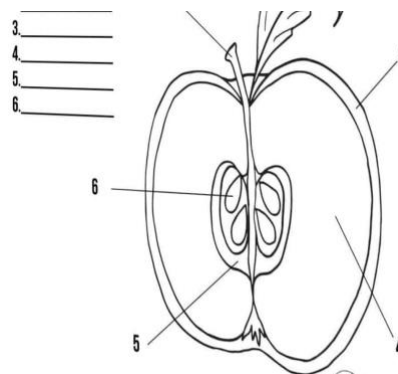
IV. Fill in the blanks.

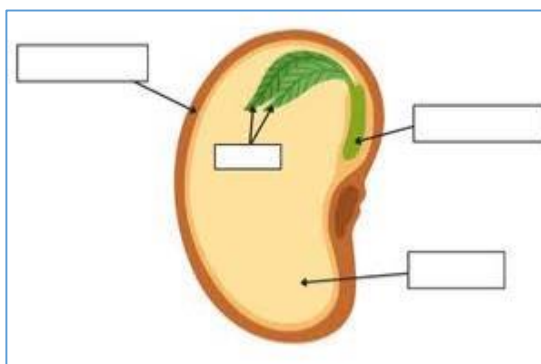
- The fertilized ovule develops into _____.
- Cotyledon is also called _____.
- The female reproductive part of a flower is called the _____.
- The seed contains the _____ that develops into a baby plant.
- Water enters the seed through a small opening called the _____.
- The stalk of a flower is called _____.
- The tube like structure that connects the stigma to the ovary is called _____.
- The second whorl of the flower is _____.

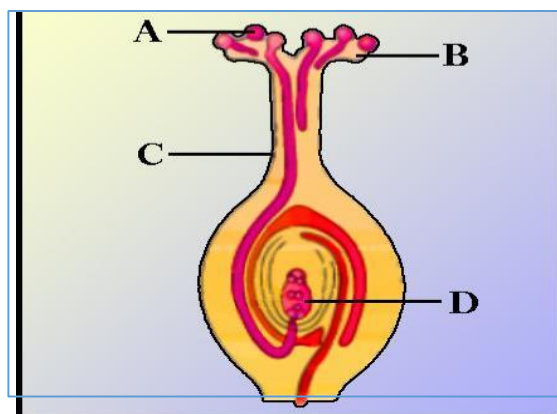
- Papaya is an example of _____ flowers.
- The part of the ovary that contains the female gametes are called _____.
- The zygote undergoes division and develops into an _____.
- _____ is a nutritive tissue and provides food to the embryo.
- Fruits developed from parts other than the ovary are called _____.
- In walnut, the hard shell is the _____ that encloses the walnut seed.
- The optimum temperature range for the seed germination is between _____ and _____.

Question 2

I. Picture based questions

1. 
- Identify the picture
 - Label the parts 3, 4, 5 and 6 in the picture
 - Write the function of all the parts in the picture.
 - Define simple fruits

2. 
- Identify the types of seed in the picture.
 - Observe the picture and identify which part forms the shoot system after germination.
 - Give another example of this type of seed.

3. 
- Identify the parts A,B,C and D in the picture.
 - How does the male gamete reach the ovule?
 - What are the parts of a flower that changes into fruits and seeds after fertilization?

II. Draw and label the following.

i) Parts of a flower

ii) Parts of a fruit

iii) Parts of an androecium.

THE VIKASA SCHOOL TUTICORIN

CLASS: VI

QUESTION BANK

SUBJECT :BIOLOGY

CHAPTER:3 CELL: THE BASIC UNIT OF LIFE

I. Choose the correct option:

1. To which of the following cells can the shape of Amoeba be compared?
(A)RBC (B) WBC (C) Skin cell (D) Muscle cell
2. The structure present in animal cell but not in plant cell is this
(A) Golgi body (B) Vacuole (C) Ribosome (D) Centrosome
3. are found in plant cells.
(A) Plastids (B) Ribosomes (C) Lysosomes (D) Centrosomes
4. The supportive framework of the cell is.....
(A) RER (B) Golgibody (C) Cytoplasm (D) Organelles
5. Leeuwenhoek is known as the father of
(A) Chordata (B) Invertebrata (C) Microbiology (D) Biology
6. In..... , German biologists, Matthias Schleiden and Theodar Schwann put forward the cell theory.
(A) 1834 (b) 1838 (C) 1879 (D) 189
7. These are colourless plastids.
(A) Chromoplasts (B) Leucoplasts (C) Lysosomes (D) Vacuoles
8. The protoplasm comprises a jelly-like.....
(A) Cytoplasm (B) Nucleus (C) Chromatin (D) Mitochondria
9. The chromatin bears hereditary units called.....
(A) Chromosomes (B) Genes (C) Nucleus (D) Nuclear membrane.
10. The _____ is living and elastic in nature.
(A) Cell wall (B) Amoeba (C) Nucleus (D) All of these

II. Fill in the blanks:

1. The _____ has no ribosomes attached to it.
2. Chromatin is present in the _____.
3. The _____ bears hereditary units called genes.
4. Cells can be seen with the help of a _____
5. The _____ provides a supportive framework to the cell..

III. Write true or false:

1. The longest cell is that of an ostrich egg.
2. Centrosomes are often called the 'suicidal bags'
3. The nucleus is usually spherical in shape.
4. Nerve cells are oval in shape.
5. Ribosomes are flattened, membrane bound vesicles.

IV . State the exact location and function o the following:

- 1 Cell wall
2. Endoplasmic reticulum
3. Ribosome
4. Centrosomes
5. Chromoplasts

V. Differentiate between the following:

1. Mitochondria and Chloroplast
2. Cell wall and cell membrane
3. Lysosome and vacuole
4. Plant cell and Animal cell.
5. Unicellular and Multicellular organisms.

VI. Give reason for the following:

1. Leaves are green in colour
2. Mitochondria are known as the powerhouse of a cell
3. The nucleus is called the brain of the stem

V. Name the following:

1. Colourless plastids
2. Green plastids
3. The plastids contain red orange and yellow pigments
4. The cell organelle also called the "power house of the cell."
5. Stacked, flattened, membrane bound vesicles.

VI. Short answer question:

- 1 . What does the cell theory state?
2. Where are genes located and what is its function?
3. Differentiate between plant cell and animal cell.
4. What are plastids?
5. Differentiate between unicellular and multicellular organisms
6. Who proposed cell theory?
7. Leaves are green in colour. Give reason.
8. What is the function of nucleus?
9. What is protoplasm?
10. What are plastids?

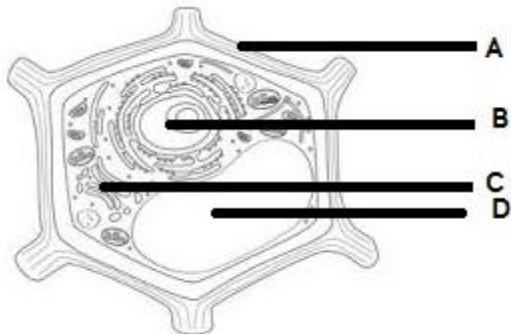
VII. Draw and label the parts of the following diagrams:

1. Plant cell
2. Animal cell
3. Nucleus

4 Mitochondria

5. Microscope

VIII. **Observe the figure and answer the following questions:**



1. State whether it is a plant cell or an animal cell. Give two reasons in support of your answer.

2. Label the parts numbered 1-4.

3. Write the function of A.s

IX. Assertion And Reasoning questions:

1. Assertion: Cells can be seen with the help of a microscope.

2. Reason: A microscope is an instrument used to see very small objects that are not visible to the naked eye.

a) Both A and R are True b) Both A and R are False

c) A is true and R is False d) A is false and R is True

2. Assertion: Plastids found in Animal cells

Reason: They are bound by a double membrane and can be oval, spherical, or discoid in shape.

a) Both A and R are True b) Both A and R are False

c) A is true and R is False d) A is false and R is True

2. Assertion: There are 23 pairs of chromosomes in the nucleus of a human cell.

Reason: The last pair of chromosomes are called sex chromosomes, represented by XY in males and XX in females

- a) Both A and R are True b) Both A and R are False
- c) A is true and R is False d) A is false and R is True

3. Assertion: All animals and plants are made up of cells.

Reason: Tissues are the basic units of structure and function in all organisms.

4. Assertion: A cell is the basic structural and functional unit of life.

Reason: All organisms are composed of one or more cells

- a) Both A and R are True b) Both A and R are False
- c) A is true and R is False d) A is false and R is True

5. Assertion: Antonie Van Leeuwenhoek was one of the pioneers who developed a compound microscope.

Reason: Leeuwenhoek is known as the father of microbiology.

- a) Both A and R are True b) Both A and R are False
- c) A is true and R is False d) A is false and R is True